

FYS-STK3155/4155 AUG 31



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$$y = f(x) + \varepsilon$$

$\varepsilon \sim N(0, \sigma^2)$

non-stochastic
function

$$P_{\varepsilon}(x) = \frac{1}{\sqrt{2\pi}\sigma^2}$$

$$x \exp\left(-\frac{x^2}{2\sigma^2}\right)$$

$$E[\varepsilon] = 0$$

$$\text{Var}[\varepsilon] = \sigma^2$$

Expectation values

$$E[x^n] = \int_{x \in D} x^n P_X(x) dx$$

$$\left(\text{or } \sum_{x_i \in D} x_i^n P_X(x_i) \right)$$

Mean value

$$\mu_X = E[x] = \langle x \rangle =$$

$$\int_{x \in D} dx P_X(x) x$$

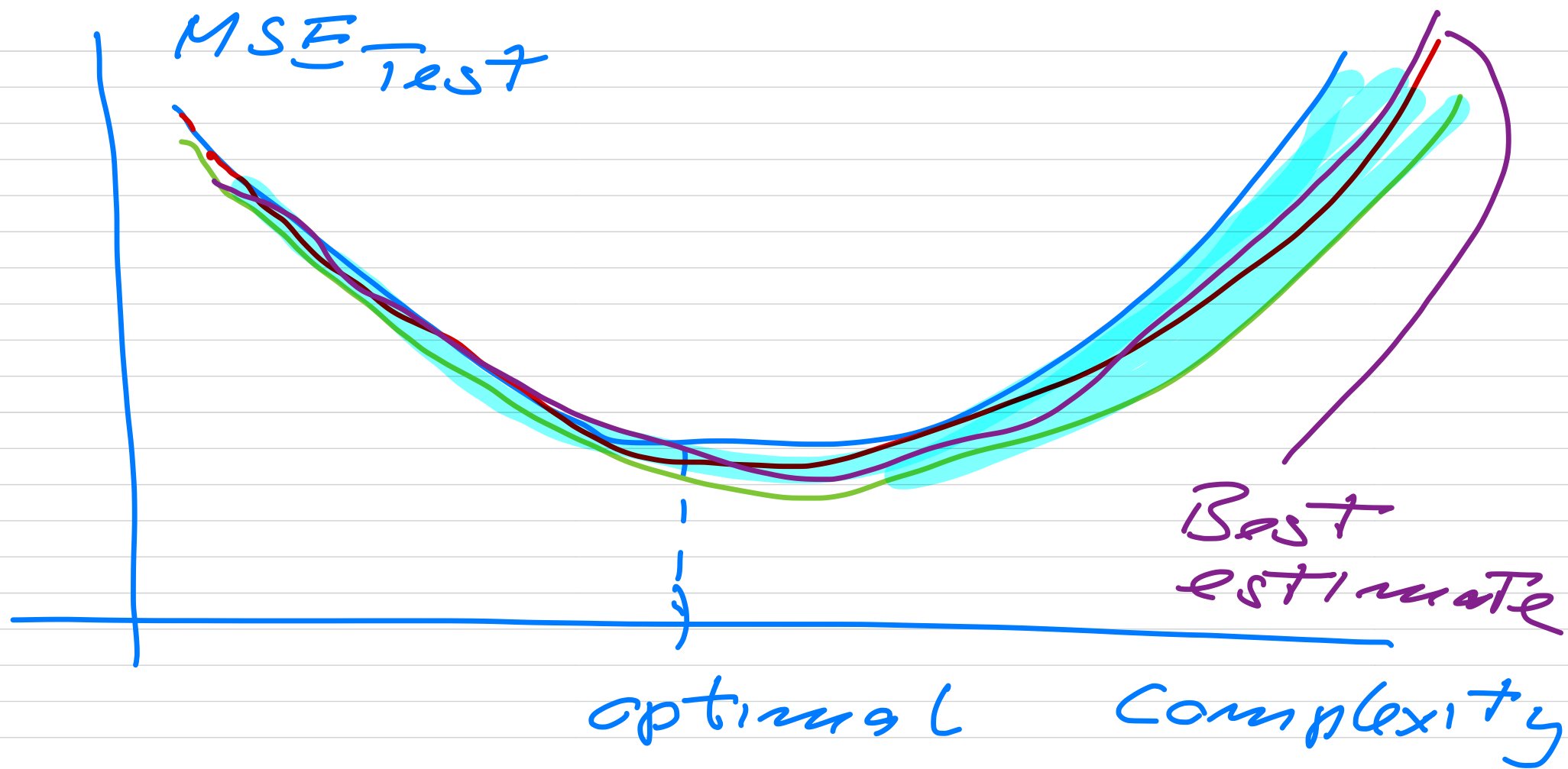
Sample mean

$$\bar{\mu}_x = \frac{1}{n} \sum_{i=0}^{n-1} x_i \quad (\neq \mu_x)$$

MSE

$$\frac{1}{n} \sum_{i=0}^{n-1} (y_i - \hat{y}_i)^2$$

$$= \mathbb{E}[(\vec{y} - \hat{\vec{y}})^2]$$



Best estimate from
"experimental" resampling
methods.

$$\text{var}[x] = E[x^2] - \mu_x^2$$

$$= \int_{x \in D} P_x(x) dx (x - \mu_x)^2$$

sample variance

$$\hat{\sigma}_x^2 = \frac{1}{n-1} \sum_{i=0}^{n-1} (x_i - \bar{\mu}_x)^2$$

$$y_i \approx \sum_{j=0}^{p-1} x_{ij} \Theta_j + \varepsilon_i$$

$$f(x_i) \approx \sum_{j=0}^{p-1} x_{ij} \Theta_j = x_i * \Theta$$

$$E[y_i] = \underbrace{E[x_i * \Theta]}_{\substack{= \\ 0}} + E[\varepsilon_i] \quad ?$$

$$\int_{y \in D} P_y(s) dy = \underline{1}$$

$$E[X_{i*} \in] = \int_{y \in D} P_y(s) dy$$

$$\times X_{i*} \in$$

$$= X_{i*} \in \underbrace{\int_{y \in D} P_y(s) dy}_{= \underline{1}}$$

$$E[y_i] = X_{i*} \theta$$

$$E[y] = X \theta$$

$$\text{var}[y_i] = \sigma^2 = \sigma^2$$

$$y_i \sim \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(y_i - X_{i*}\theta)^2}{2\sigma^2} \right]$$

i.i.d. = independent & identically distributed.

$$P(y_i | \theta) = P_i = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(y_i - x_i^T \theta)^2}{2\sigma^2}\right]$$

i.i.d. :

$$P(y | \theta) = \prod_{i=0}^{n-1} P_i$$

$$\hat{\theta} = \arg \max_{\theta \in \mathbb{R}^p} P(y | \theta)$$

$$C(\theta) = -\log P$$

$$= -\sum_{i=0}^{n-1} \log P_i$$

$$= \frac{n}{2} \log (2\pi \sigma^2)$$

$$+ \sum_{i=0}^{n-1} \frac{(y_i - \sum_{j=0}^{p-1} x_{ij} \theta_j)^2}{2\sigma^2}$$

$$\nabla_{\theta} C(\theta) = 0 = -\frac{1}{\sigma^2} X^T (y - X\theta)$$

$$\Rightarrow \hat{\theta} = (X^T X)^{-1} X^T y = \hat{\theta}_{OLS}$$

Bias-variance Tradeoff

$$MSE = \frac{1}{n} \sum_{i=0}^{n-1} (y_i - \tilde{y}_i)^2$$

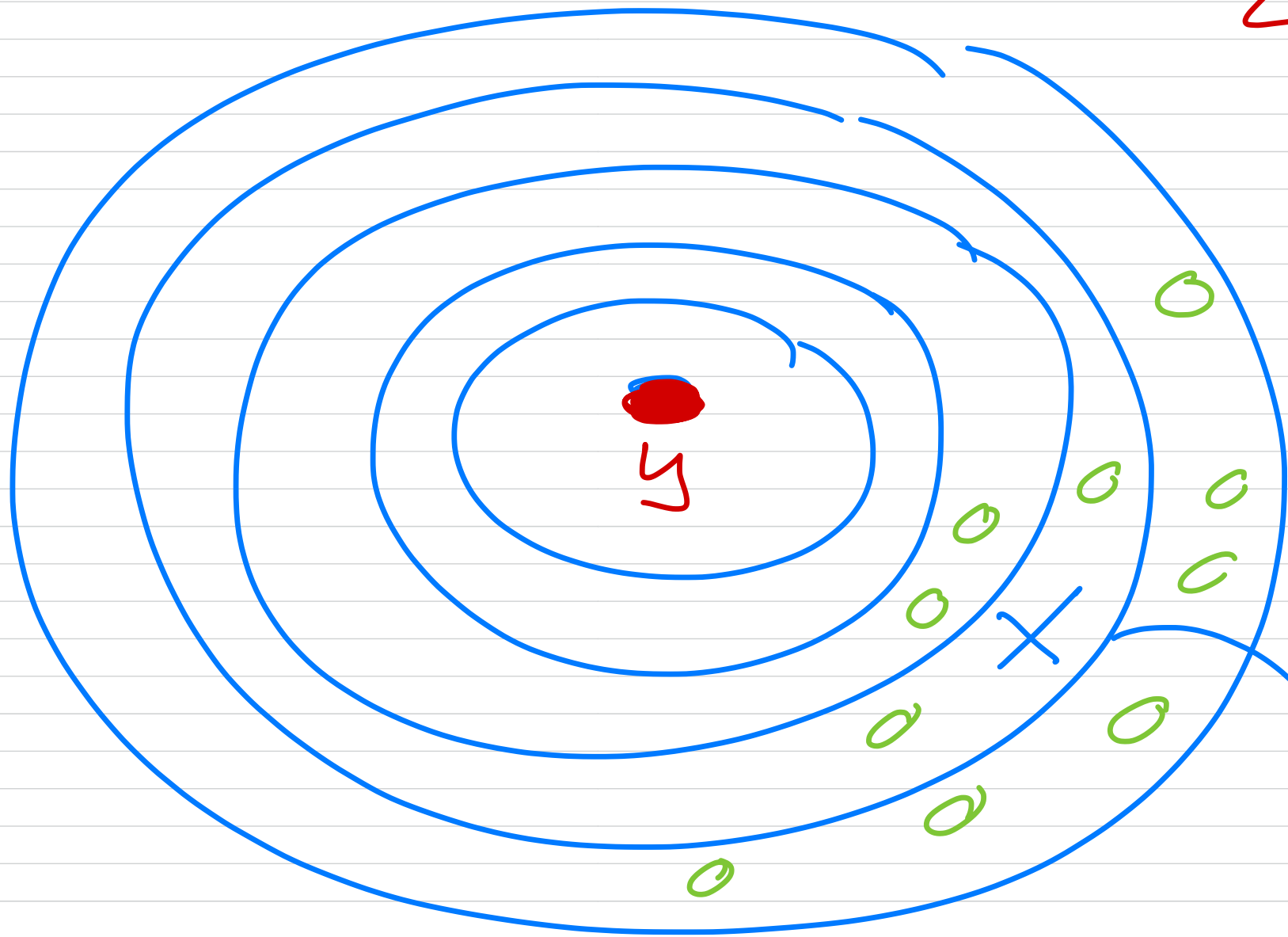
$$= E[(y - \tilde{y})^2]$$

$$(y = f(x) + \varepsilon)$$

$$= E[(\overset{\text{true } y}{f} - E[\tilde{y}])^2]$$

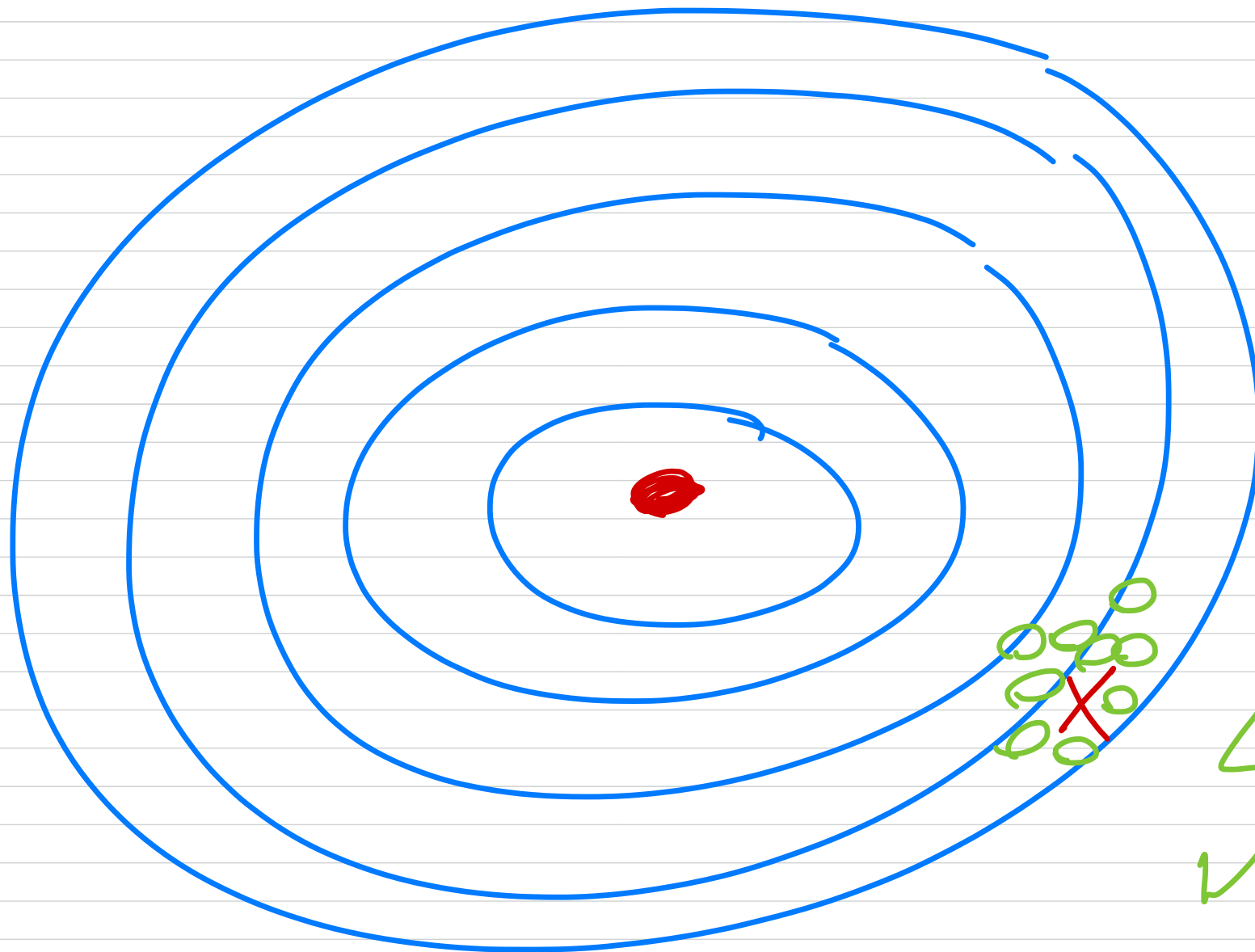
Bias

$$+ \underset{\text{variance}}{\text{Var}[\tilde{y}]} + \sigma^2$$



Large
bias
High
variance

$E[y]$



High
Bias

Low
Variance

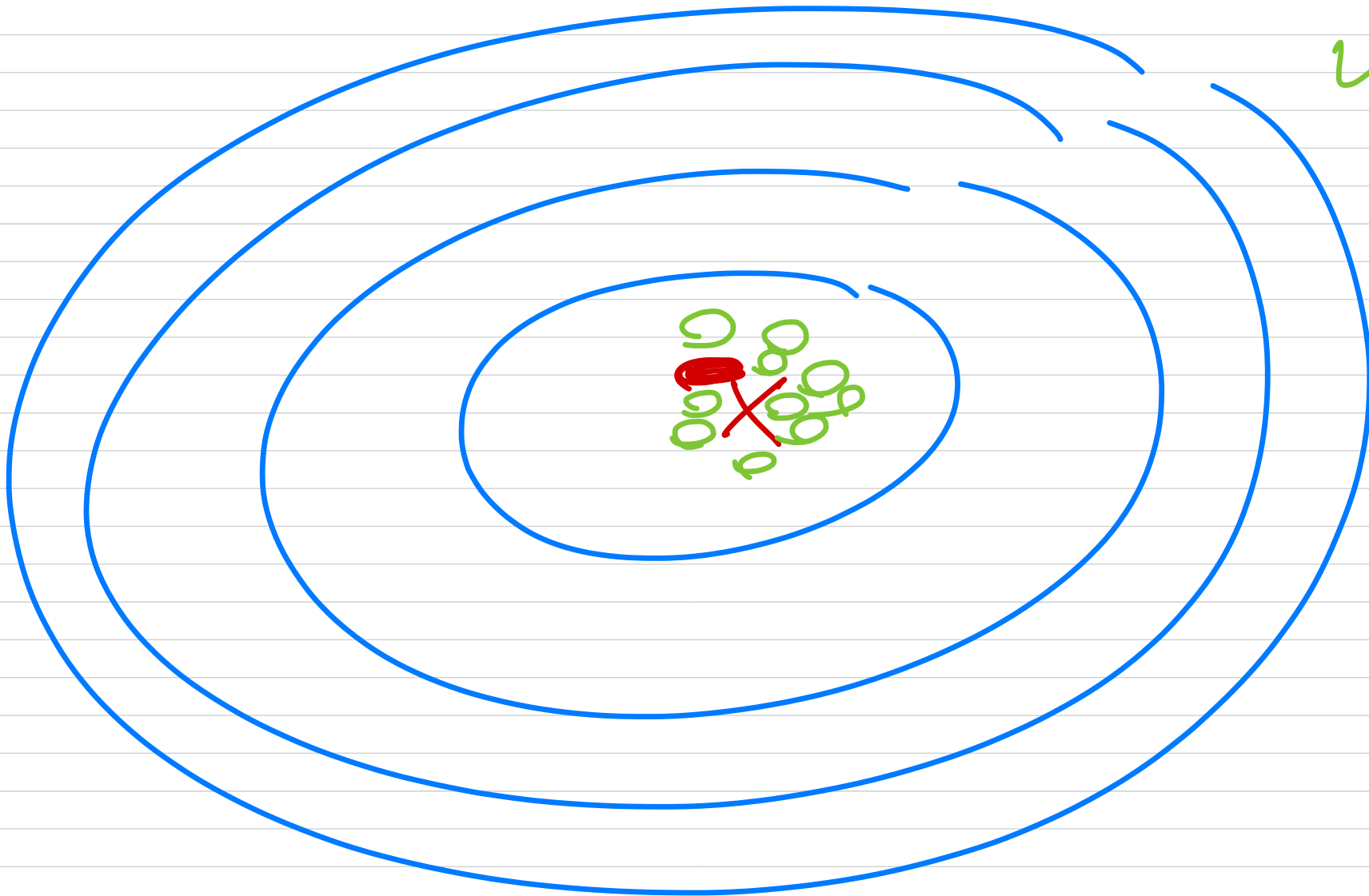
ideal

Low

Variance

Low

bias



Expression for BV-Tradeoff

$$E[(y - \hat{y})^2] =$$

$$E[y^2] - 2E[y\hat{y}] + E[\hat{y}^2]$$

$$(var[\hat{y}] = E[\hat{y}^2] - (E[\hat{y}])^2)$$

$$= E[y^2] - 2E[y\hat{y}] + var[\hat{y}] + (E[\hat{y}])^2$$

$$y = f + \varepsilon$$

$$\begin{aligned}
 E[Y^2] &= E[(f + \varepsilon)^2] \\
 &= E[f^2] + 2E[f \cdot \varepsilon] \\
 &\quad + \underbrace{E[\varepsilon^2]}_{\sigma^2}
 \end{aligned}$$

$$E[f \cdot \varepsilon] = f \underbrace{E[\varepsilon]}_0$$

$$E[Y^2] = E[f^2] + \sigma^2 \approx 0$$

$$E[y^2] = E[(f + \varepsilon)^2]$$

$$= \underbrace{E[f^2]}_{\int E[y]} + \underbrace{E[\varepsilon^2]}_{\substack{= 0 \\ E[\varepsilon] E[y]}}$$

$$\int E[y]$$

$$\underbrace{E[\varepsilon]}_{=0} E[y]$$

collecting

$$E[f^2] - 2 \int E[y] + (E[y])^2 + \text{var}[y] + \sigma^2$$

$$= \underbrace{E[(f - E[\tilde{y}])^2]}_{\text{Bias}} + \underbrace{\text{var}[\tilde{y}]}_{\text{variance}} + \sigma^2$$

f is unknown; small
cheat $f \approx y$

$$E[(y - E[\tilde{y}])^2] + \text{var}[\tilde{y}] + \sigma^2$$

Resampling Methods

— Bootstrap

$$D = [z_0, z_1, \dots, z_{n-1}]$$

sample mean

$$\bar{\mu}_z = \frac{1}{n} \sum_{i=0}^{n-1} z_i$$

$$\bar{\sigma}_z^2 = \frac{1}{n-1} \sum_{i=0}^{n-1} (z_i - \bar{\mu}_z)^2$$

$$\bar{\mu}_z \pm \bar{\sigma}_z$$

Central limit theorem $\bar{\sigma}_z \rightarrow \frac{\bar{\sigma}_z}{\sqrt{n-1}}$
m-experiments

$$D' = [z_0' \ z_1' \ \dots \ z_{n-1}']$$

reshuffle data randomly
with replacement

$$D = [1, 2, 3, 4, 5]$$

$$D' = [2, 2, 4, 1, 3]$$

⋮

B - times / repetitions,

$$\bar{\mu}_B = \frac{1}{B} \sum_{k=1}^B \bar{\mu}_k$$

$$s_B^2 = \frac{1}{B-1} \sum_{k=1}^B (\bar{\mu}_k - \bar{\mu}_B)^2$$